

ORIGINAL ARTICLE

Assessment of antifouling efficacy of polyhedral oligomeric silsesquioxane based poly (urea–urethane–imide) hybrid membranes

P. Ajit Walter¹, T. Muthukumar² and B.S.R. Reddy¹

¹ Industrial Chemistry Laboratory, CSIR-Central Leather Research Institute, Chennai, India

² Department of Biotechnology, Indian Institute of Technology-Madras, Chennai, India

Significance and Impact of the Study: The antifouling performance has been evaluated for the polymeric membranes against two bacteria (*Staphylococcus aureus* (ATCC 6538)) (*Escherichia coli* (ATCC 8739)). The polymeric membranes were incorporated with imide moiety to improve thermal stability of the polymeric materials. The synthesized polymeric membranes have shown good morphological properties for better antifouling activities. This study found that these membranes are capable of preventing micro-organisms besides offering excellent bio-fouling resistance.

Keywords

anti-adhesion, anti-biofouling, hydrophobic, polyhedral oligomeric silsesquioxane, silicone polymers.

Correspondence

Boreddy Siva Rami Reddy, Industrial Chemistry Laboratory, CSIR-Central Leather Research Institute, Adyar, Chennai 600 020, India.
E-mail: induchem2000@yahoo.com

2015/0233: received 5 February 2015, revised 27 May 2015 and accepted 27 May 2015

doi:10.1111/lam.12457

Abstract

A series of polyhedral oligomeric silsesquioxane (POSS) based poly(urea–urethane–imide) (PUUI-POSS) membranes were synthesized by varying the proportions of imide using 2,4-toluene diisocyanate (TDI) and bis (aminopropyl) terminated polydimethylsiloxane (PDMS). The molecular structures of poly(urea–urethane–imide)s were characterized by Attenuated total reflection Fourier transform infrared (ATR-FTIR) spectroscopic technique. Incorporation of imide domain and its influence on surface roughness was investigated by atomic force microscopy (AFM). Hydrophobicity of polymeric membrane surfaces was determined by contact angle measurement. The thermal properties of the polymers were studied by thermogravimetric analysis. The antimicrobial activities and inhibition of bacterial attachment of these polymeric membranes were studied on Gram-positive (*Staphylococcus aureus*) and Gram-negative bacteria (*Escherichia coli*) by the disc-diffusion method.

Introduction

The accumulation of organisms such as mussel, acorn barnacles and algae onto submerged man-made structures poses a severe problem. The initial stage of bio-fouling begins at the time of immersion and consists of a complex community of bacteria, diatoms and protozoa that attach to a layer of organic matter. Once this biofilm was formed, macro organisms such as barnacles, algae and sponges can easily attach to the membrane surfaces. A layer of these higher ordered fouling organisms on ship's hull can decrease its speed and increase fuel consumption (Brady 2001; Stupak *et al.* 2003). Therefore, preventing the attachment of bacteria during the early stages of

biofilm formation could greatly reduce the subsequent biofouling problems (Bosselmann 1996; McBain *et al.* 2000; Mah and O'Toole 2001).

Most of the antifoulants were toxic; a metal-based compound tributyltin (TBT) was traditionally used to keep the unwanted organisms away. A more serious issue occurred while using metal-based compounds like TBT is that they reduce fouling by killing the attached organisms and also harmful to other nontarget organisms as it is nonselective. These harmful effects include abnormal growth, reproductive failure, sterility and premature death of various marine species at concentrations even below 1 ng l⁻¹ (Bryan *et al.* 1996; Alzieu 1998; Callow and Callow 2000). TBT is not the only antifouling material that poses environmental threats.

By using other toxic metal-based antifouling paints such as zinc, chromium and cadmium poses serious environmental problems even at very low concentrations and have been restricted (Bennett 1996). Therefore, there is a need to investigate suitable nontoxic alternatives. One alternative is to create foul-release (FR) coatings that rely on materials with low surface energy or coating ablation. Silicones and fluorinated polymers are the most widely used materials for this application (Whelan and Regan 2006; Callow and Callow 2011; Li *et al.* 2014a,b). Many silicones are transparent and could be used mainly to protect optical surfaces from fouling. It should be noted that silicones do actually get weakly fouled and are more properly termed 'fouling release coatings' (Swain *et al.* 1998; Kavanagh *et al.* 2001; Whelan and Regan 2006). The bond between the foulant and the surface is so weak that the fouling material is either displaced by shear forces.

In the present study, one of the main general approaches have been made to control bio-fouling is material design. This was divided broadly into 'anti-adhesion' approaches to reduce initial stage micro molecular adsorption or attachment of organisms and 'antimicrobial' approaches which attack, disperse or suppress the activity of attached organisms. By keeping these objectives in mind we have synthesized a series of poly(urea-urethane-imide) membranes containing silicones such as polydimethylsiloxane (PDMS) and polyhedral oligomeric silsesquioxane (POSS). We have assessed the efficiency of antifouling properties of these polymeric materials by attachment study.

Results and discussion

ATR-FTIR spectroscopy of PUUI membranes

The chemical structures of the PUU and poly(urea-urethane-imide) (PUUI) membranes were characterized by ATR-FTIR. The disappearance of NCO stretching vibrations $2200\text{--}2400\text{ cm}^{-1}$ confirm the completion of the reaction (Li *et al.* 2011) (Fig. S1) provides the broad band of N-H stretching vibration of urethane/urea observed at 3319 cm^{-1} (Sadeghi *et al.* 2010). Free carbonyl and hydrogen bonded carbonyl absorption of the urethane/urea groups in the PUUI membranes are assigned in the region from 1600 to 1800 cm^{-1} . In PUU and PUUI membranes, which can form medium to strong hydrogen bonding has its carbonyl peak maximum appeared at 1660 cm^{-1} for urethane, whereas, the urea can form very strong bidentate hydrogen bonding shows a peak maximum at 1638 cm^{-1} , this is much lower than the ester or urethane (Lee *et al.* 1987; Teo *et al.* 1997; Yilgor *et al.* 2006; Jena *et al.* 2007). Furthermore, C–N–H bending in

the urethane linkages can be observed around 1556 cm^{-1} (Li *et al.* 2011). Moreover, The characteristic bands of imide bonds that is symmetrical stretching of imide at 1720 cm^{-1} and C–N stretching of imide appeared at 1387 and 1256 cm^{-1} , confirming that imide moiety was incorporated in to polyurea-urethane backbone (Mishra *et al.* 2006; Tsai *et al.* 2008; Gnanasekaran *et al.* 2013).

Atomic force microscopy of PUUI membranes

The effect of POSS and imide content on the surface morphology of PUUI membranes were studied by atomic force microscopy (AFM). The three-dimensional topographical images were given in (Fig. S2). The mean roughness (R_a) and root mean square (R_q) values were presented in (Table 1). The surface roughness is the important factor for the wettability of the surface (Kostov *et al.* 2012). The roughness values (R_q) for the PUU and PUUI-20% samples are 12.4 and 37.71 nm respectively. Accordingly, the surface free energies are 32.30 and 39.20 mN m^{-1} for PUU and PUUI-20% respectively. The surface roughness of the film increases as the siloxane cross-linking decreases while increasing the imide content. This property attributes the wettability of the films. The contact angle also decreases with increasing R_q surface roughness value. Furthermore, the contact angle of the surface was $>90^\circ$ for all the polymeric membranes, indicating the hydrophobicity of the surface. This means that the surface was covered with more siloxane groups than hard segments. Therefore, this could be attributed to the existence of noncompatible phases. Similar correlation of surface roughness to the phase separation has been reported by Viville *et al.* (Viville *et al.* 2001) and Lin *et al.* (Lin *et al.* 2000) for tetramethyl bisphenol A polycarbonate and polystyrene blends.

Contact angle measurements of PUUI membranes

The contact angle measurements provide information about the surface energies of the membranes. The surface energy of polymeric membrane can be calculated indirectly by measuring equilibrium contact angles of some standard liquids, such as water and higher alkanes, on the air side of membrane surface. The polar and the dispersion factors were calculated using Young and Fowks equation (1).

$$\gamma_{LV}(1 + \cos\theta) = 2(\gamma_L^d \gamma_S^d)^{1/2} + (\gamma_L^p \gamma_S^p)^{1/2} \quad (1)$$

where γ_{LV} is the interfacial tension at liquid/air interface; γ_L^d and γ_S^d are the dispersion factors of the sample for liquid and solid respectively; γ_L^p and γ_S^p are the polar factors of the liquid and sample; θ is the contact angle between the sample and liquid/air interface. The total

Table 1 Contact angles, surface energies and surface roughness of PUU and PUUI- polyhedral oligomeric silsesquioxane membranes

Sample	$\theta_{\text{H}_2\text{O}}$	$\theta_{\text{C}_{17}\text{H}_{36}}$	$\gamma_{\text{S}}^{\text{d}}$ (mN m ⁻¹)	$\gamma_{\text{S}}^{\text{p}}$ (mN m ⁻¹)	γ_{SV} (mN m ⁻¹)	Surface roughness (nm)	
						R_{a}	R_{q}
PUU	99.76	24.23	25.12	7.18	32.30	9.89	12.4
PUUI- 10	95.15	26.95	24.59	11.01	35.60	16.71	20.72
PUUI-15	92.48	28.04	24.36	13.36	37.72	23.73	30.70
PUUI- 20	90.55	29.89	23.96	15.24	39.20	32.14	37.71

$\theta_{\text{H}_2\text{O}}$: Contact angle of water; $\theta_{\text{C}_{17}\text{H}_{36}}$: Contact angle of heptadecane; $\gamma_{\text{S}}^{\text{d}}$: Dispersion factors of the sample; $\gamma_{\text{S}}^{\text{p}}$: Polar factors of sample; γ_{SV} : Total surface energy of the samples; PUUI, poly(urea–urethane–imide).

surface energy γ_{SV} of the samples were estimated using the relationship (2).

$$\gamma_{\text{SV}} = \gamma_{\text{S}}^{\text{d},\text{p}} \quad (2)$$

The most of PDMS based polymeric membrane was found to have higher contact angle in water (i.e. above 80°) (Park and Lee 2004). (Table 1) presented the static water contact angles of PUU and PUUI-POSS membranes. After incorporation of 10, 15 and 20 wt % of imide onto the PUU membrane, the contact angles of water remarkably decreased from 99.76° for PUU membrane to 95.15, 92.48 and 90.55 respectively. This shows that hydrophobicity decreases due to increasing the amount of imide content of the prepared poly(urea–urethane–imide). This observation indicates that the hydrophobic nature of PDMS/POSS plays a major role in influencing contact angles than surface roughness. The results of AFM and contact angle measurements complement each other. The surface free energy (γ_{SV}) correspondingly increased from 32.30 mN m⁻¹ to 35.60, 37.72 and 39.20 mN m⁻¹. The surface free energy of a solid surface is an index of intermolecular or interfacial attractive forces (Li *et al.* 2014b). These low surface free energy properties of these membranes would be expected to minimize the adhesion propensity and facilitate the detachment of foulants from membrane surface (Krishnan *et al.* 2008; Magin *et al.* 2010).

Assessing antimicrobial activity and inhibition of bacterial attachment over PUUI membranes

Bacterial culture

Two bacterial cultures were used in this study including one Gram-positive (*Staphylococcus aureus* (ATCC 6538)) and one Gram-negative bacteria (*Escherichia coli* (ATCC 8739)). All the cultures were sub-cultured periodically and maintained on nutrient agar (NA) medium at room temperature (30 ± 2°C) for further experiments. The cultures were grown over night in NA at 37°C. The washed cells were suspended in NA broth, and the optical density

(OD) was adjusted to 0.1, which corresponds to 10⁸ CFU (colony forming units)/ml⁻¹ at 600nm.

Culture of fibroblast cells

The mouse embryonic fibroblasts (NIH 3T3) cells line was obtained from National Centre for Cell Sciences (NCCS), Pune, India and the culture were maintained in Dulbecco's Modified Eagles Medium (DMEM) with 10% foetal bovine serum (FBS) supplemented with penicillin (120 U ml⁻¹), streptomycin (75 mg ml⁻¹), gentamycin (160 mg ml⁻¹) and amphotericin B (3 mg ml⁻¹) at 37°C humidified with 5% CO₂.

Disc-diffusion assay

To examine the antimicrobial susceptibility to *Staph. aureus* and *E. coli*, various poly(urea–urethane–imide) membranes (PUU, PUUI-10%, PUUI-15% and PUUI-20%) were cut into 1 × 1 cm, and they were evaluated using the Kirby-Bauer disc-diffusion method (Pal *et al.* 2007). Briefly, Muller–Hinton agars were sterilized and poured onto petri plates and allowed to solidify under laminar airflow. The antimicrobial activity was carried out by agar disc-diffusion method; about 100 µl (10⁸ CFU ml⁻¹) of each bacterial culture was spread on the agar surface using sterile glass spreader. The polymeric membranes were placed on the agar plate and then incubated for 24 h at 35 ± 2°C. After 24 h the antibacterial activity was evaluated by measuring the diameter of zone of inhibition against the test organism. Streptomycin disc was used as a positive control.

Generally, nonfouling materials can significantly reduce the initial attachment and delay the colonization of microbes on surfaces and they cannot kill or inhibit the growth of pathogenic bacteria cells once bacteria are attached to surfaces. The antimicrobial strategy is another method for preventing bacterial colonization on surfaces (Cheng *et al.* 2010). Quaternary ammonium compounds (QACs) are extensively used as antimicrobial agents because of their broad antimicrobial properties (Chen *et al.* 2005; Cheng *et al.* 2007, 2010). The antimicrobial

activities of the polymeric membranes are shown in (Fig. 1B). The activities of the composites were measured using a Gram-positive and Gram-negative bacteria. All the polymeric membranes were found to have moderate activities. The difference of antimicrobial and antifouling properties of PUU and PUUI membranes may also be attributed by the surface property changes of the membranes. According to the contact angle results obtained in this study, PUU membrane was found to be hydrophobic; the hydrophobic feature of the membranes could limit the entry of the aqueous bacteria suspension into the inner part of the membrane. Incorporation of the hard segment (imide moiety) on to PUU membranes there was distinct reduction in the contact angle values and dramatically enhanced surface hydrophilicity. Therefore, the bacteria may accumulate on the surface of the membrane. These accumulated bacteria can be easily killed by the presence of urea moieties present on the surface of the membranes and not by the presence of PDMS and POSS. Herein these silicone materials act as a nonfouling agent, these agents could not be killed or deter the bacteria, they can reduce bacterial attachment and colonization on the surface.

Antifouling properties of PUUI membranes

For the experiments, *Staph. aureus* and *E. coli* were grown in tryptic soy agar (TSA) and broth (TSB). The ability of POSS incorporated PUUI membranes to control bacterial adhesion on the surface of the membranes was performed by scanning electron microscopy (SEM) and backlight dye (live and dead cells). Specifically, sterile polymeric nanocomposite material of 1×1 cm was immersed in a conical flask containing 25 ml of bacterial suspension (0.5 MacFarland) grown in TSB broth for 24 h at an agitation speed of 120 rev min^{-1} at 37°C . Later the polymer materials were collected using sterile forceps and washed several times with phosphate-buffered saline (PBS, pH 7.4) to remove loosely adherent bacterial cells, followed by ultra-sonication for 10 min with 1 min intervals and transfer of the material into test tubes with 10 ml of Ringer's solution. This was then vortexed for 10s to remove adherent cells. This procedure was repeated three times for each sample. For SEM analysis, after the washing step with PBS (pH 7.4), the samples were fixed with 2.5% glutaraldehyde in 0.1 mol l^{-1} PBS buffer (pH 7.4) at room temperature for 30 min, dehydrated through graded ethanol (25, 50, 75, 90 and 100%) followed by critical point drying and gold coated by sputtering. The samples were then examined with a scanning electron microscope.

The percentage of cell reduction was determined using live/dead Backlights Bacterial Viability Kit (Invitrogen, Molecular Probes, Darmstadt, Germany). The kit consists

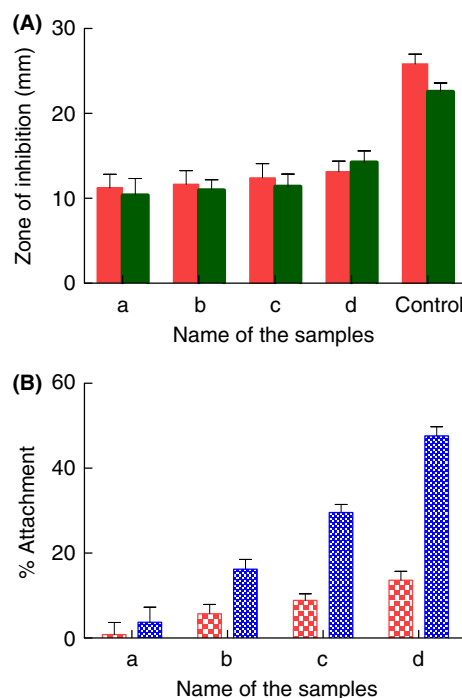


Figure 1 (A) % of bacterial attachment over (a) PUU (b) PUUI-10%, (c) PUUI-15% and (d) PUUI-20% membranes. ■ *Staphylococcus aureus*; ■ *Escherichia coli*. (B) Zone of inhibition of (a) PUU, (b) poly (urea-urethane-imide) (PUUI)-10%, (c) PUUI-15% and (d) PUUI-20%; ▨ *Staphylococcus aureus*; ▨ *Escherichia coli*.

of SYTO9 and propidium iodide dyes. These two dyes differ in their ability to stain bacterial cells. The SYTO9 stains the live as well as the dead cells as green and the propidium iodide stains only the dead cells as red, by penetrating into the damaged membrane of the cells (Boulos *et al.* 1999). Hence, live cells fluorescence green was read at 530 nm and dead cells fluorescence red was read at 630 nm with an excitation wavelength at 485 nm in a fluorometer (FP-6500, Jasco spectrofluorometer, Mumbai, India). The green to red ratio (*G/R*-ratio) was given as a measure of live to dead cells.

Bacterial attachment to polymeric material is a complex process to explain and is influenced by several parameters, such as properties of material, bacterial cell and liquid environment. Many researchers have found that in *in vitro* environment, compare to the hydrophilic surface the hydrophobic surfaces showed comparatively less bacterial adhesion (Quirynen *et al.* 1989; Everaert *et al.* 1997). Our results also support the reported literatures that a significant decrease in bacterial attachment in the hydrophobic material and it reveals that the material possess antibacterial activity. The inhibition of bacterial attachment over PUU and PUUI polymer membranes were given in (Fig. 1A) for *E. coli* and *Staph. aureus*. The percentage of

attachment of *Staph. aureus* was very less compared to *E. coli* which might be correlated with the antimicrobial study by disc-diffusion method which also showed higher activity for Gram-positive bacteria *Staph. aureus*. Distinct increase in the bacterial attachment with an increase in the concentration of the imide hard segment in the polymer nanocomposites and also revealed that there was a major decrease in the bacterial attachment and count with respect to the concentration of PDMS in polymeric materials (Guo *et al.* 2011). The polymeric membranes showed 0.81, 4.61, 8.87 and 13.67% of the attachment for *Staph. aureus* and 3.70, 21.12, 32.16 and 47.62% of the attachment for *E. coli* respectively for PUU, PUUI-10%, PUUI-15% and PUUI-20% polymer composites. Robin Jude *et al.* (Michael *et al.* 2014) have also reported about 40 and 32% bacterial attachment for *E. coli* and *Staph. aureus* for silver nanoparticles and clusters in ZnO matrix. The increase in the bacterial attachment could be also correlated with surface roughness of the polymer composites. Attachment of bacteria on the membranes surface was attributed to the enhancement of interaction between the micro-organisms with an increase in surface roughness. That means, the micro-organisms preferentially accumulate at the valleys of the rough membrane surface. As a result, valleys become blocked and fouling becomes more severe for the rougher membrane surface. Similar type of observation was also reported by

Elimelech *et al.* (1997) and Vrijenhoek *et al.* (2001). The inhibition of bacterial attachment over PUUI membranes is very clear for *Staph. aureus* (Fig. 2a,b) and *E. coli* (Fig. 2c,d) from the SEM images. The SEM results clearly corroborates with the contact angle measurements indicating that the PUUI membranes tested were useful for exploitation on a commercial scale for antifouling applications.

Determination of cell cytotoxicity by MTT

Cell viability in terms of metabolic activity was determined calorimetrically by evaluating the uptake of 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) by the cells (Russell *et al.* 2000). The sterile poly-(urea-urethane-imide) material were seeded with fibroblasts in the 12 well plates at the density from 8×10^3 cells/well to 7×10^6 cells/well were incubated for 48 h at 37°C humidified with 5% CO₂. After the incubation the wells were flushed with serum free DMEM and later centrifuged at 900 g for 15 min, the cells were collected and 0.5% MTT reagents were added. The mixture was incubated in dark for 4 h at 37°C with serum free medium. The reagents were aspirated and 100% DMSO were added for cell lysis. The lysate were used to read at 570 nm for cell viability assay on a spectrophotometer (DSH-UV751GD Spectrophotometer, China). The optical

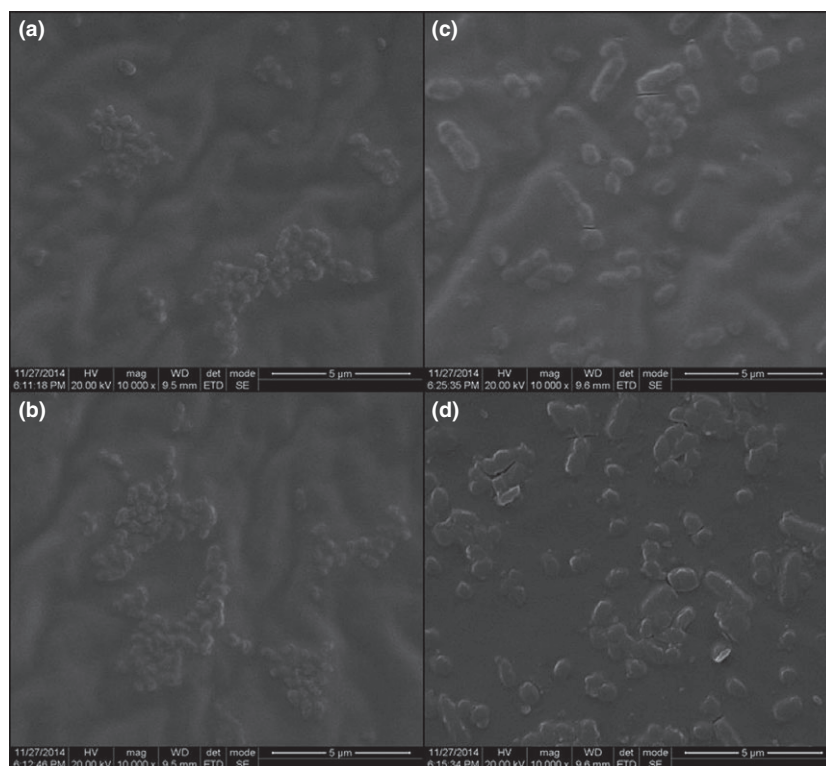


Figure 2 SEM images of (i) *Staph. aureus* attachment on PUU (a) and poly(urea-urethane-imide) (PUUI)-10% (b) membranes; (ii) *E. coli* attachment on PUU (c) and PUUI-10% (d) membranes.

density values were calculated and the results were presented in mean $\pm$ standard deviation (SD). All samples were performed in triplicates. Percentage cell viability was calculated by

$$\% \text{ Viability} = \frac{\text{Mean absorbance of sample}}{\text{Mean absorbance of control}} \times 100$$

Cytotoxicity of the prepared polymeric membranes was investigated using MTT by seeding NIH 3T3 Fibroblast cells onto them. The percentage of cell viability on the membranes was given in (Fig. S3). The obtained results showed all the materials were biocompatible showing more than 89% cell viability using 3T3 cells. Compared with the control (polystyrene surface) all the cells grown on the prepared polymeric membrane were found to be uniform with normal morphology. Extensive fibroblast growth and attachment were seen on the control and no observable cell death or inhibition of growth was noticed in treated wells (prepared material). More than 93% cell viability was seen on PUU sample and more than 91% cell viability was seen on both PUUI-10% and PUUI-15%, the PUUI-20% sample showed 89% cell viability. Researchers have reported that like other fillers POSS contains organic constituents on their surface that makes them compatible with many polymers (Kuo and Chang 2011). Some of them have reported that, dispersing POSS nanoparticles into the polymer increases the strength, rigidity, modulus, heat discharge and viscosity of the material (Chan *et al.* 2007; Zhou *et al.* 2008). However, the biocompatibility of the polymer material and the material moderate hydrophilicity improved the cell-substrate affinity that provides good support for the cell adhesion. This biocompatibility results clearly suggest that the material can be used for the biomedical applications.

In summary, we have synthesized PUU and a series of PUUI hybrid membranes having various percentages of imide functionalities in the polymeric backbone. Structural characteristics of the polymeric membranes were characterized by ATR-FTIR analysis by confirming the presence of urea, urethane and imide moieties in the polymer. Observations from TGA showed that the higher imide (PUUI-20%) shows higher thermal decomposition temperature compared to the other polymeric composites. Furthermore, the membranes were characterized for their surface properties such as membrane hydrophilicity/hydrophobicity and membrane surface roughness *via* the contact angle measurements and AFM respectively. Obviously, the PUU membrane is more hydrophobic and smoother than other membranes. It was also noted that both properties have great impacts on the fouling rate of membranes. The results indicated that higher hydrophobicity of membrane contributes to better resistance to

fouling of membrane. AFM topography also clearly revealed that lower surface roughness leads to higher fouling resistance. PUU and PUUI-10% membranes have shown the best antifouling property due to their higher degree of PDMS and POSS surface coverage.

Materials and methods

Heptacyclopentyl tricycloheptasiloxane triol (Cy-POSS) was synthesized in our laboratory and the details were given in our previous publication (Gnanasekaran *et al.* 2011). 2, 4-Toluene diisocyanate (TDI; $\bar{M}_n = 172 \text{ gmol}^{-1}$), bis (aminopropyl) terminated PDMS ($\bar{M}_n = 2500 \text{ gmol}^{-1}$) and benzophenone-3,3',4,4'-tetracarboxylic dianhydride ($\bar{M}_n = 322 \text{ gmol}^{-1}$) were all purchased from Aldrich and used as received. Dibutyltin dilaurate (DBTDL, Aldrich, 95%) and tetrahydrofuran (THF, Aldrich) were distilled before use. All other chemicals used were of analytical grade and were used as received.

The attenuated total reflectance Fourier transforms infrared (ATR-FTIR) spectra by Perkin-Elmer spectrophotometer was used to analyse the chemical structure of the polymeric membranes. An average of 20 scans was performed for all samples at a resolution of 2 cm^{-1} .

Contact angle measurements were carried out at room temperature by sessile drop method. The surface free energy of PU membranes were calculated by measuring contact angle measurements in a double distilled water and n-heptadecane. The contact angle was measured at five different locations for all samples and the average value was taken to obtain meaningful measurement.

The thermal stability of the polymers were determined using NETZSCH STA 449F3 JUPITER thermal analysers by taking 10 mg of samples at a heating rate of 10°C per minute under nitrogen atmospheres.

Surface morphology of the polymers was studied using a Nanoscope III AFM instrument and the imaging was done in contact mode at room temperature in air. The membranes were cut into small pieces and placed on a grid. The grid was covered using the commercial tip of Si_3N_4 provided by digital instruments. Cantilever length was kept at $10 \mu\text{m}$ with a spring constant of 0.12 Nm^{-1} .

Synthesis

Synthesis of anhydride terminated prepolyimide (PPI)

Benzophenone-3, 3', 4, 4'-tetracarboxylic dianhydride in 6 ml of THF was placed in a flask equipped with a nitrogen inlet and stirred using magnetic stirrer until clear solution was obtained. To the clear solution, 2, 4-toluene diisocyanate in 4 ml of THF was added slowly under stirring. The reaction mixture was refluxed at 90°C in 6 h by

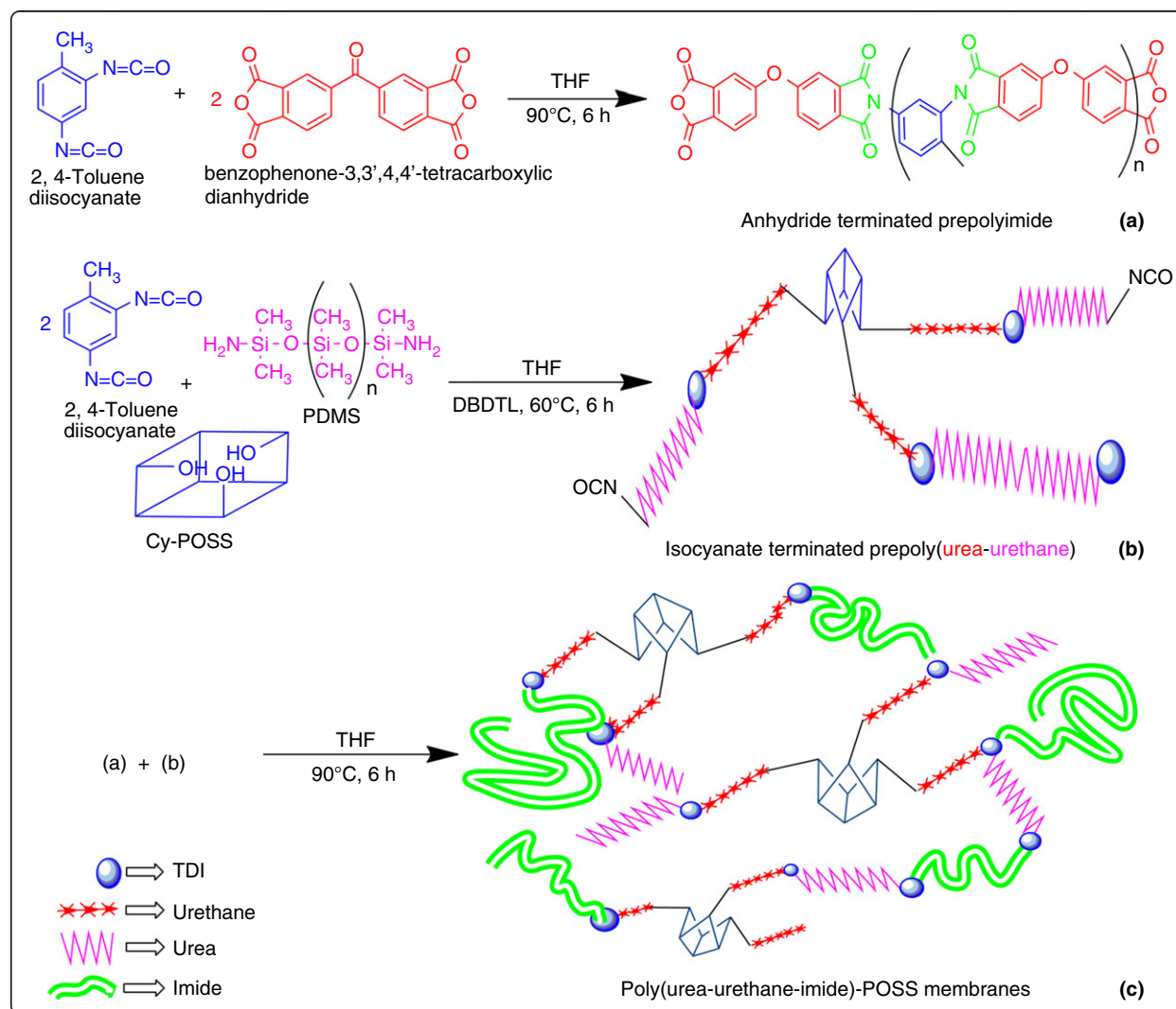


Figure 3 Synthesis of poly(urea-urethane-imide)-polyhedral oligomeric silsesquioxane membranes. TDI; Urethane; Urea; Imide.

Table 2 Chemical composition of PUUI membranes

Sample	Prepolyurea-urethane (PUU) (g)			Prepolyimide (PPI) (g)	
	TDI	PDMS	Cy-POSS	TDI	DA
PUU	0.200	2.775	0.1	—	—
PUUI- 10%	0.330	2.275	0.1	0.070	0.257
PUUI- 15%	0.313	2.150	0.1	0.095	0.354
PUUI- 20%	0.295	2.025	0.1	0.139	0.515

PDMS, polydimethylsiloxane; POSS, polyhedral oligomeric silsesquioxane; PUUI, poly(urea-urethane-imide)

connecting a spiral condenser. The synthetic route was given in (Fig. 3a). The chemical compositions of the synthesized PPIs were given in (Table 2).

Synthesis of NCO terminated prepolyurea-urethane (PUU)

The NCO terminated prepolyurea-urethane was prepared by reacting PDMS and TDI in a 50 ml two-necked flask. PDMS and Cy-POSS were reacted with TDI in 5 ml of THF, followed by adding two drops of DBTDL as a catalyst at 60°C for 6 h. The synthetic route was given in (Fig. 3b). The chemical distributions of synthesized PUU were given in (Table 2).

Synthesis of poly(urea-urethane-imide) hybrid membranes (PUUI)

The PUUI membranes were prepared by varying the ratios of prepolyimide and prepolyurea-urethane in the ratio of 0/100, 10/90, 15/85 and 20/80 respectively. THF was used as a solvent for refluxion by using spiral condenser at 90°C for 6 h. The resulting viscous solution was

transferred into a teflon coated petri dish and was kept at room temperature for 6 h. All the membranes were kept in hot air oven at 90°C for 7 h. The model reaction of this polymer was shown in (Fig. 3c).

Acknowledgements

One of the authors P.A is thankful to the Department of Science and Technology for the financial support (No: SR/S1/PC-45/2011). P.A is also thankful to Dr. D. Gnanasekaran, Dielectric Material Division, Central Power Research Institute, Sir C.V Raman Road, Bangalore-560 080, India, for providing me the starting material Cy-POSS.

Conflict of Interest

All authors have no conflict of interest.

References

- Alzieu, C. (1998) Tributyltin: case study of a chronic contaminant in the coastal environment. *Ocean Coast Manag* **40**, 23–36.
- Bennett, R. (1996) Industrial manufacture and applications of tributyltin compounds. *R F Bennett, 'Tributyltin: Case Study of an Environmental Contaminant'* (ed. De Mora, S.J.), Cambridge University Press (1996), 21–6, WSCA Abstract.
- Bosselmann, K. (1996) Environmental law and tributyltin in the environment. *Tributyltin: Case Study of an Environmental Contaminant*. pp. 237–263. Cambridge: Cambridge University Press.
- Boulos, L., Prevost, M., Barbeau, B., Coallier, J. and Desjardins, R. (1999) LIVE/DEAD® BacLight™: application of a new rapid staining method for direct enumeration of viable and total bacteria in drinking water. *J Microbiol Methods* **37**, 77–86.
- Brady, R.F. Jr (2001) A fracture mechanical analysis of fouling release from nontoxic antifouling coatings. *Prog Org Coat* **43**, 188–192.
- Bryan, P.J., Rittschof, D. and McClintock, J.B. (1996) Bioactivity of echinoderm ethanolic body-wall extracts: an assessment of marine bacterial attachment and macroinvertebrate larval settlement. *J Exp Mar Biol Ecol* **196**, 79–96.
- Callow, M.E. and Callow, J. (2000) Substratum location and zoospore behaviour in the fouling alga *Enteromorpha*. *Biofouling* **15**, 49–56.
- Callow, J.A. and Callow, M.E. (2011) Trends in the development of environmentally friendly fouling-resistant marine coatings. *Nat Commun* **2**, 244.
- Chan, E.R., Striolo, A., McCabe, C., Cummings, P.T. and Glotzer, S.C. (2007) Coarse-grained force field for simulating polymer-tethered silsesquioxane self-assembly in solution. *J Chem Phys* **127**, 114102.
- Chen, S., Zheng, J., Li, L. and Jiang, S. (2005) Strong resistance of phosphorylcholine self-assembled monolayers to protein adsorption: insights into nonfouling properties of zwitterionic materials. *J Am Chem Soc* **127**, 14473–14478.
- Cheng, G., Zhang, Z., Chen, S., Bryers, J.D. and Jiang, S. (2007) Inhibition of bacterial adhesion and biofilm formation on zwitterionic surfaces. *Biomaterials* **28**, 4192–4199.
- Cheng, G., Xue, H., Li, G. and Jiang, S. (2010) Integrated antimicrobial and nonfouling hydrogels to inhibit the growth of planktonic bacterial cells and keep the surface clean. *Langmuir* **26**, 10425–10428.
- Elimelech, M., Zhu, X., Childress, A.E. and Hong, S. (1997) Role of membrane surface morphology in colloidal fouling of cellulose acetate and composite aromatic polyamide reverse osmosis membranes. *J Membr Sci* **127**, 101–109.
- Everaert, E., Mahieu, H., Chung, R.W., Verkerke, G., Van der Mei, H. and Busscher, H. (1997) A new method for *in vivo* evaluation of biofilms on surface-modified silicone rubber voice prostheses. *Eur Arch Otorhinolaryngol* **254**, 261–263.
- Gnanasekaran, D., Madhavan, K., Tsibouklis, J. and Reddy, B. (2011) Ring opening metathesis polymerization of polyoctahedral oligomeric silsesquioxanes (POSS) incorporated oxanorbornene-5, 6-dicarboximide: synthesis, characterization, and surface morphology of copolymers. *Aust J Chem* **64**, 309–315.
- Gnanasekaran, D., Ajit Walter, P., Asha Parveen, A. and Reddy, B. (2013) Polyhedral oligomeric silsesquioxane-based fluoroimide-containing poly (urethane-imide) hybrid membranes: synthesis, characterization and gas-transport properties. *Sep Purif Technol* **111**, 108–118.
- Guo, W., Tang, X., Xu, J., Wang, X., Chen, Y., Yu, F. and Pei, M. (2011) Synthesis, characterization, and property of amphiphilic fluorinated abc-type triblock copolymers. *J Polym Sci, Part A: Polym Chem* **49**, 1528–1534.
- Jena, K.K., Chattopadhyay, D. and Raju, K. (2007) Synthesis and characterization of hyperbranched polyurethane-urea coatings. *Eur Polym J* **43**, 1825–1837.
- Kavanagh, C.J., Schultz, M.P., Swain, G.W., Stein, J., Truby, K. and Wood, C.D. (2001) Variation in adhesion strength of *Balanus eburneus*, *crassostrea virginica* and hydroids *dianthus* to fouling-release coatings. *Biofouling* **17**, 155–167.
- Kostov, K., Dos Santos, A., Nascente, P., Kayama, M., Mota, R. and Algatti, M. (2012) Surface modification of siloxane containing polyurethane polymer by dielectric barrier discharge at atmospheric pressure. *J Appl Polym Sci* **125**, 4121–4127.
- Krishnan, S., Weinman, C.J. and Ober, C.K. (2008) Advances in polymers for anti-biofouling surfaces. *J Mater Chem* **18**, 3405–3413.

- Kuo, S.-W. and Chang, F.-C. (2011) POSS related polymer nanocomposites. *Prog Polym Sci* **36**, 1649–1696.
- Lee, H.S., Wang, Y.K. and Hsu, S.L. (1987) Spectroscopic analysis of phase separation behavior of model polyurethanes. *Macromolecules* **20**, 2089–2095.
- Li, H., Freeman, B.D. and Ekiner, O.M. (2011) Gas permeation properties of poly (urethane-urea)s containing different polyethers. *J Membr Sci* **369**, 49–58.
- Li, Y., Su, Y., Zhao, X., He, X., Zhang, R., Zhao, J., Fan, X. and Jiang, Z. (2014a) Antifouling, high-flux nanofiltration membranes enabled by dual functional polydopamine. *ACS Appl Mater Interfaces* **6**, 5548–5557.
- Li, Y., Su, Y., Zhao, X., Zhang, R., Zhao, J., Fan, X. and Jiang, Z. (2014b) Surface fluorination of polyamide nanofiltration membrane for enhanced antifouling property. *J Membr Sci* **455**, 15–23.
- Lin, Q., Sooriyakumaran, R. and Huang, W.-S. (2000) Toward controlled resist line-edge roughness: material origin of line-edge roughness in chemically amplified positive-tone resists. In *Advances in Resist Technology and Processing XVII* ed. F.M. Houlihan pp. 230–239. Santa Clara, CA: International Society for Optics and Photonics.
- Magin, C.M., Cooper, S.P. and Brennan, A.B. (2010) Non-toxic antifouling strategies. *Mater Today* **13**, 36–44.
- Mah, T.-F.C. and O'Toole, G.A. (2001) Mechanisms of biofilm resistance to antimicrobial agents. *Trends Microbiol* **9**, 34–39.
- McBain, A.J., Allison, D. and Gilbert, P. (2000) Emerging strategies for the chemical treatment of microbial biofilms. *Biotechnol Genet Eng Rev* **17**, 267–280.
- Michael, R.J.V., Sambandam, B., Muthukumar, T., Umapathy, M.J. and Manoharan, P.T. (2014) Spectroscopic dimensions of silver nanoparticles and clusters in ZnO matrix and their role in bioinspired antifouling and photocatalysis. *Phys Chem Chem Phys* **16**, 8541–8555.
- Mishra, A.K., Chattopadhyay, D., Sreedhar, B. and Raju, K. (2006) FT-IR and XPS studies of polyurethane-urea-imide coatings. *Prog Org Coat* **55**, 231–243.
- Pal, S., Tak, Y.K. and Song, J.M. (2007) Does the antibacterial activity of silver nanoparticles depend on the shape of the nanoparticle? A study of the gram-negative bacterium *Escherichia coli*. *Appl Environ Microbiol* **73**, 1712–1720.
- Park, Y.W. and Lee, D.S. (2004) Surface and electrical characteristics of poly (amide imide)-poly (dimethylsiloxane) nanocomposites. *J Appl Polym Sci* **93**, 342–347.
- Quirynen, M., Marechal, M., Busscher, H., Weerkamp, A., Arends, J., Darius, P. and van Steenberghe, D. (1989) The influence of surface free-energy on planimetric plaque growth in man. *J Dent Res* **68**, 796–799.
- Russell, R.C.G., Williams, N.S. and Bulstrode, C.J.K. ed. (2000) *Bailey & Love's Short Practice of Surgery*. 23rd edn. pp. 1032–1033. London: Arnold.
- Sadeghi, M., Semsarzadeh, M.A., Barikani, M. and Ghalei, B. (2010) The effect of urethane and urea content on the gas permeation properties of poly (urethane-urea) membranes. *J Membr Sci* **354**, 40–47.
- Stupak, M.E., García, M.T. and Pérez, M.C. (2003) Non-toxic alternative compounds for marine antifouling paints. *Int Biodeterior Biodegradation* **52**, 49–52.
- Swain, G.W., Nelson, W.G. and Preedeeekanit, S. (1998) The influence of biofouling adhesion and biotic disturbance on the development of fouling communities on non-toxic surfaces. *Biofouling* **12**, 257–269.
- Teo, L.-S., Chen, C.-Y. and Kuo, J.-F. (1997) Fourier transform infrared spectroscopy study on effects of temperature on hydrogen bonding in amine-containing polyurethanes and poly (urethane-urea)s. *Macromolecules* **30**, 1793–1799.
- Tsai, M.-H., Huang, S.-L., Liu, S.-J., Chen, C.-J., Chen, P.-J. and Chen, S.-H. (2008) Synthesis and properties of poly (urethane-imide) interpenetrating network membranes. *Desalination* **233**, 191–200.
- Viville, P., Biscarini, F., Bredas, J. and Lazzaroni, R. (2001) Scaling aspects of the kinetics of thermally induced phase separation in bisphenol A polycarbonate/poly (methyl methacrylate) blends. *J Phys Chem B* **105**, 7499–7507.
- Vrijenhoek, E.M., Hong, S. and Elimelech, M. (2001) Influence of membrane surface properties on initial rate of colloidal fouling of reverse osmosis and nanofiltration membranes. *J Membr Sci* **188**, 115–128.
- Whelan, A. and Regan, F. (2006) Antifouling strategies for marine and riverine sensors. *J Environ Monit* **8**, 880–886.
- Yilgor, I., Yilgor, E., Guler, I.G., Ward, T.C. and Wilkes, G.L. (2006) FTIR investigation of the influence of diisocyanate symmetry on the morphology development in model segmented polyurethanes. *Polym J* **47**, 4105–4114.
- Zhou, Z., Cui, L., Zhang, Y., Zhang, Y. and Yin, N. (2008) Preparation and properties of POSS grafted polypropylene by reactive blending. *Eur Polym J* **44**, 3057–3066.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Figure S1 ATR-FTIR spectra of PUUI membranes (a) PUU; (b) PUUI-10% and (c) PUUI-20%.

Figure S2 Three-dimensional AFM images of (a) PUU, (b) PUUI-10%, (c) PUUI-15% and (d) PUUI-20% membranes.

Figure S3 Determination of cell cytotoxicity of (a) PUU, (b) PUUI-10%, (c) PUUI-15% and (d) PUUI-20% membranes.